AUTOMATED BEAM MECHANICS SOLVER: COMPUTATION AND VISUALIZATION OF SHEAR FORCE AND BENDING MOMENT DIAGRAMS (SFD & BMD)

Project Report

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For The Course

21CSO352T – Python Programming



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EXECUTIVE SUMMARY

This project report presents the systematic design, theoretical formulation, and incremental development of a high-performance Python computational engine for the structural analysis of statically determinate beams. The software calculates external reaction forces and moments, evaluates continuous Shear Force Diagrams (SFD) and Bending Moment Diagrams (BMD), locates critical zero-shear crossing points, computes global peak bending moments, and identifies points of contraflexure. The engine supports a rich array of boundary conditions (Simply Supported, Overhanging, and Cantilever beams) and arbitrary combinations of transverse loading—including concentrated point loads (vertical or inclined), uniformly distributed loads (UDL), uniformly varying loads (UVL with linear trapezoidal/triangular profiles), and concentrated point couples.

The project evolved across four major milestone versions (v1, v2, v3, and v4). Starting from a rudimentary point-load CLI script, the architecture matured into an industrial-grade engineering application featuring persistent multi-table CSV ingestion, bidirectional coordinate referencing (distances from either the left or right beam boundary), an interactive review and modification loop, and high-resolution 3-tier publication graphics (Free Body Diagram, SFD, and BMD). The final version (sfd_bmd_engine_4.py) was rigorously benchmarked and validated against Example 4.7 from the classic textbook 'Strength of Materials' by R. K. Rajput. The software's numerical predictions achieve exact 0.00% agreement with analytical closed-form solutions, validating both its mathematical rigor and programming robustness.

> **[PROJECT METADATA]**
> *Project Scope: Academic Software Development for 21CSO352T (Python Programming).*
> *Department: Mechanical Engineering, College of Engineering & Technology, SRM Institute of Science and Technology.*
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> *Final Submitted Source Code: sfd_bmd_engine_4.py*

1. INTRODUCTION & THEORETICAL FOUNDATIONS

In structural engineering, civil engineering, and mechanical machine design, beams are fundamental load-bearing members subject to transverse forces, distributed pressures, and external bending couples. The internal stresses generated across any cross-section are directly characterized by two key quantities: Shear Force (V) and Bending Moment (M). Designing safe and cost-effective structural elements requires plotting these quantities as continuous functions of longitudinal position x, yielding the Shear Force Diagram (SFD) and Bending Moment Diagram (BMD).

1.1 Governing Differential Equations of Beam Mechanics

The behavior of a beam in static equilibrium is governed by the classical differential equations relating distributed load intensity w(x), internal transverse shear force V(x), and internal bending moment M(x):

- **1. Load-Shear Differential Relation:** $dV(x) / dx = -w(x)$ [The rate of change of shear force equals the negative of load intensity]

- **2. Shear-Moment Differential Relation:** $dM(x) / dx = V(x)$ [The rate of change of bending moment equals the transverse shear force]
- **3. Second-Order Relation:** $d^2M(x) / dx^2 = -w(x)$ [Curvature and second derivative relationship]

From calculus, when a continuous function's first derivative equals zero, the function reaches a local extremum. Because $dM/dx = V(x)$, the bending moment $M(x)$ attains its maximum or minimum values precisely at locations where the shear force crosses zero ($V(x) = 0$). Identifying these zero-shear points with sub-millimeter precision is essential, as the peak bending moment dictates the required elastic section modulus ($Z = M_{max} / \sigma_{allowable}$) to prevent structural failure or plastic yielding.

1.2 Static Equilibrium Conditions

For statically determinate coplanar beam systems, the unknown support reactions are uniquely determined from the equations of rigid-body static equilibrium in two dimensions:

- **Transverse Equilibrium:** $\Sigma F_y = 0$ --> Total Upward Support Reactions = Total Downward Applied Loads
- **Rotational Equilibrium:** $\Sigma M_z(0) = 0$ --> Algebraic sum of all moments about any arbitrary reference point (e.g., left origin x=0) must vanish.

For simply supported or overhanging beams with two supports (pinned and roller) located at x1 and x2, taking moments about x1 yields:

- **Reaction at Support 2:** $R2 = [\Sigma (P_i * x_i) + \Sigma \int w(x)*x\, dx + \Sigma M_{applied} - Total_Force * x1] / (x2 - x1)$
- **Reaction at Support 1:** $R1 = Total_Force - R2$

For cantilever beams clamped rigidly at support x_s, the wall provides a single vertical reaction $R_y = Total_Force$ and a clamping counter-moment $M_{fixed} = -(Total_Moment - Total_Force * x_s)$.

1.3 Mechanical Sign Conventions

Consistent sign conventions are critical to avoid catastrophic errors in sign and diagram interpretation. The software strictly adheres to international civil and mechanical engineering standards:

- **Shear Force Convention:** Shear Force (V): Positive (+) when an upward force acts to the left of the cutting section or downward force to the right.
- **Bending Moment Convention:** Bending Moment (M): Sagging bending moments (which create concave upward curvature, inducing compression in top fibers and tension in bottom fibers) are POSITIVE (+). Hogging bending moments (convex upward curvature, inducing tension in top fibers) are NEGATIVE (-).
- **Concentrated Moments:** Applied Moments: Clockwise external couples are treated as positive (+), Counter-Clockwise as negative (-).
- **Contraflexure Points:** Points of Contraflexure: Locations along the beam where the bending moment changes sign (M = 0), representing zero-curvature inflection points where bending stress reverses direction.

2. PROJECT GOAL & ARCHITECTURAL OVERVIEW

The overarching goal of this project is to engineer an accurate, fully automated, and highly resilient Python computational engine capable of analyzing arbitrary statically determinate beam configurations under complex multi-load scenarios. Manual calculation of SFD and BMD diagrams is time-consuming, prone to algebraic errors, and difficult to visualize. This software solves this problem by integrating rigorous numerical mechanics, robust error handling, multi-source data ingestion, and synchronized multi-panel plotting.

2.1 Key Functional Requirements

- **1. Determinacy Support:** Support all determinate boundary conditions: Cantilever (fixed wall left or right) and Simply Supported / Overhanging beams.
- **2. Load Diversity:** Accommodate point loads (vertical or inclined at arbitrary angles), UDL spans, UVL trapezoidal/triangular profiles, and concentrated couples.
- **3. Dual-Coordinate Ingestion:** Eliminate manual coordinate shifting by enabling the user to specify distances either from the left end (x = 0) or the right end (x = L).
- **4. Dense Numerical Grid:** Formulate closed-form reactions and apply singularity function integration over 5,000 discrete points for sub-millimeter precision.
- **5. Critical Point Analytics:** Calculate zero-shear crossings via linear interpolation, extract global peak bending moments, and mark contraflexure points.
- **6. Persistence & Export:** Enable persistent data storage using multi-section CSV files and export publication-ready 300 DPI graphics.
- **7. Interactive Fault Tolerance:** Provide an interactive verification loop that allows reviewing and modifying inputs before executing the solver.

2.2 High-Level System Architecture

The software architecture follows a modular decoupled pipeline divided into six cohesive functional subsystems:

Subsystem Module	Primary Python Functions	Role & Engineering Functionality
1. Input & Coordinate Transformer	get_float(), get_choice(), get_position()	Robust validation, type safety, range checking, and dual-reference coordinate mapping (left/right).
2. Interactive Load Collectors	setup_beam_interactive(), collect_point_loads(), collect_udls(), collect_uvls(), collect_moments()	Step-by-step modular collection of beam geometry, supports, inclined forces, distributed loads, and couples.
3. Review & Verification Loop	review_and_verify()	Interactive pre-flight inspection of beam configuration; allows editing, adding loads, or restarting before calculation.
4. CSV Persistence Engine	load_from_csv(), save_to_csv()	Structured multi-table CSV serialization and deserialization with directory-agnostic path handling via SCRIPT_DIR.

5. Equilibrium & Singularity Solver	solve_reactions(), calculate_sfd_bmd()	Analytical reaction solver via static equilibrium equations, followed by dense Macaulay/singularity numerical integration across 5,000 points.
6. Critical Analytics & Visualizer	find_critical_points(), create_figure()	Zero-crossing bisection, maximum moment detection, contraflexure identification, and 3-panel synchronized Matplotlib rendering.

3. EVOLUTIONARY DEVELOPMENT ACROSS VERSIONS

Developing this software followed a structured agile methodology. Rather than creating a single monolithic program, the engine was systematically designed and refined across four distinct milestone versions: Version 1 (sfd_bmd_engine_1.py), Version 2 (sfd_bmd_engine_2.py), Version 3 (sfd_bmd_engine_3.py), and the final production-ready Version 4 (sfd_bmd_engine_4.py). The following sections chronicle the engineering objectives, architectural upgrades, limitations encountered, and code advancements for each iteration.

3.1 Version 1 (sfd_bmd_engine_1.py) – Initial Computational Prototype

Goal of Version 1: Establish a minimal working proof-of-concept that can compute support reactions and plot basic SFD and BMD curves for a simply supported beam or cantilever beam carrying concentrated vertical point loads.

Progress & Implementation Details: Version 1 successfully implemented safe terminal input prompts (get_float, get_choice), an analytical equilibrium reaction calculator for simply supported and cantilever boundary conditions, and a basic numerical evaluation of shear force and bending moment over 5,000 points. A standard Matplotlib window with two vertically stacked subplots (SFD and BMD) was used to visualize the result.

- **Capabilities:** Supported Beam Types: Simply Supported (pin at 0, roller at L) and Cantilever (fixed wall at 0 or L).
- Supported Loading: Strictly concentrated point loads (downward or upward).
- Visualization: Two 2D line plots for SFD and BMD with colored fill under curves.

Limitations Encountered in Version 1: Real-world structural engineering problems rarely feature only point loads. Version 1 completely lacked support for Uniformly Distributed Loads (UDL), Uniformly Varying Loads (UVL), and concentrated bending couples. Furthermore, it did not render a Free Body Diagram (FBD), could not compute exact zero-shear crossing coordinates or contraflexure points, offered no persistence mechanism, and forced the user to restart the script on any input mistake.

Code Snippet 1: Version 1 Point-Load Integration Engine

```
# sfd_bmd_engine_1.py (Excerpt: Point load calculation and basic plotting)
def calculate_sfd_bmd(length, supports, point_loads, num_points=5000):
    reactions = solve_reactions(length, supports, point_loads)
    x = np.linspace(0.0, length, num_points)
```

```
    V = np.zeros(num_points)
    M = np.zeros(num_points)

    # Superposition of reactions
    for name, val in reactions.items():
        x_r = float(name.split('_')[1])
        mask = x >= x_r
        if name.startswith('R_'):
            V[mask] += val
            M[mask] += val * (x[mask] - x_r)
        elif name.startswith('M_'):
            M[mask] += val

    # Superposition of point loads only
    for pl in point_loads:
        mask = x >= pl['pos']
        V[mask] -= pl['mag']
        M[mask] -= pl['mag'] * (x[mask] - pl['pos'])

    return x, V, M, reactions
```

3.2 Version 2 (sfd_bmd_engine_2.py) – Extended Load Spectrum & 3-Panel Visualizer

Goal of Version 2: Expand the mechanical load handling to cover distributed loads (UDL, UVL) and couples, implement analytical critical point detection (zero shear, peak moment, contraflexure), and upgrade visualization into a synchronized 3-panel display (Free Body Diagram, SFD, and BMD).

Progress & Implementation Details: Version 2 introduced calculus-based integration routines for UDL spans and UVL linear profiles. For UVLs with varying intensity from w1 to w2, exact centroid integration formulas were derived and incorporated into both the reaction solver and the slice-by-slice Macaulay integrator. In addition, Version 2 introduced find_critical_points() to search for sign changes in V(x) and M(x). The visualization was transformed into a professional 3-panel plot using matplotlib.patches (Rectangles for UDLs, Polygons for UVLs, angled arrows for forces, and curved arcs for moments).

- **Upgrades:** New Loading Capabilities: Uniformly Distributed Loads (UDL), Uniformly Varying Loads (UVL), and Concentrated Couples.
- Critical Point Tracking: Numerical detection of zero-shear locations and points of contraflexure (M = 0).
- 3-Tier Aligned Plots: Synchronized Free Body Diagram (FBD) directly above the SFD and BMD.

Limitations Encountered in Version 2: While Version 2 possessed complete mechanical mathematics, it remained strictly CLI-based. Entering complex problems with multiple loads required dozens of sequential terminal prompts. If a user made a single typographical error on the final load, the entire session had to be discarded and restarted. Furthermore, data could not be saved to disk, preventing batch verification or problem reuse.

Code Snippet 2: Version 2 Continuous Singularity Integration for UVL Profile

```
# sfd_bmd_engine_2.py (Excerpt: Integration of Uniformly Varying Loads - UVL)
for uvl in uvls:
    span = uvl['end'] - uvl['start']
    slope = (uvl['end_w'] - uvl['start_w']) / span
    for i, xi in enumerate(x):
        if xi > uvl['start']:
            x_eff = min(xi, uvl['end']) - uvl['start']
            w0 = uvl['start_w']
            force = w0 * x_eff + 0.5 * slope * (x_eff ** 2)
            dist = xi - uvl['start']
            moment = (w0 * dist * x_eff - 0.5 * w0 * (x_eff ** 2)
                      + 0.5 * slope * dist * (x_eff ** 2) - (1.0 / 3.0) * slope *
(x_eff ** 3))
            V[i] -= force
            M[i] -= moment
```

3.3 Version 3 (sfd_bmd_engine_3.py) – Persistent CSV Architecture & Dual Coordinate System

Goal of Version 3: Eliminate repetitive terminal data entry by establishing persistent CSV file import and export capabilities, incorporating an in-terminal user manual, and supporting dual-reference coordinate specification (distances from left or right ends).

Progress & Implementation Details: Version 3 introduced a robust CSV storage schema featuring distinct section headers: [BEAM], [SUPPORTS], [POINT_LOADS], [UDLS], [UVLS], and [MOMENTS]. It included load_from_csv() and save_to_csv() routines that allowed engineers to define complex beam problems in spreadsheet editors (such as Excel or LibreOffice Calc) and solve them instantly. A structured main menu loop was implemented, and display_manual() provided in-terminal documentation of sign conventions and CSV formatting rules.

- **Upgrades:** CSV Ingestion & Serialization: Fully reproducible problem definitions saved to standard .csv files.
- Dual-Coordinate Referencing: Distances can be entered from the left (x = 0) or the right (x = L) in both CLI and CSV.
- High-Res Image Export: Automatic export of analysis diagrams to 300 DPI PNG or vector PDF.

Limitations Encountered in Version 3: Although CSV loading functioned well, the interactive CLI input pipeline remained monolithic (a single collect_loads_interactive function), making it difficult to correct errors during interactive sessions without modifying the CSV file. Furthermore, file path resolution was sensitive to the terminal's active working directory, which caused 'file not found' errors when executed from outside the script's folder.

Code Snippet 3: Version 3 Multi-Section CSV Reader

```
# sfd_bmd_engine_3.py (Excerpt: Multi-section CSV loading logic)
def load_from_csv(filepath):
    # Reads multi-section beam configuration
    with open(filepath, mode='r', encoding='utf-8') as f:
        reader = csv.reader(f)
```

```
        for row in reader:
            if not row or not any(field.strip() for field in row):
                continue
            first_cell = row[0].strip()
            if first_cell.startswith('[') and first_cell.endswith(']'):
                current_section = first_cell.upper()
                continue
            # Parse sections: [BEAM], [SUPPORTS], [POINT_LOADS], [UDLS], [UVLS],
[MOMENTS]
            ...
```

3.4 Version 4 (sfd_bmd_engine_4.py) – Final Production-Grade Mechanics Engine

Goal of Version 4 (Final Code to be Submitted): Deliver a robust, modular, fault-tolerant engineering engine featuring interactive review/modification loops, directory-independent path resolution, sub-millimeter root-finding interpolation, and publication-ready graphics.

Progress & Final Implementation Details: Version 4 represents the complete milestone software. Key advancements include:

- **1. Modular Architecture:** Modular Load Collectors: Decomposed monolithic input into collect_point_loads(), collect_udls(), collect_uvls(), and collect_moments() for clean maintainability.
- **2. Fault-Tolerant Review Loop:** Interactive Review & Verification Loop (review_and_verify): Allows users to review all entered data in a formatted summary, add further loads, edit parameters, or restart the setup before executing the equilibrium solver.
- **3. Resilient File Handling:** Directory-Agnostic File Management: Utilizes SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__)) to ensure CSVs and plots are always loaded and saved beside the script, regardless of where the terminal session was launched.
- **4. Analytical Precision:** Sub-Grid Linear Interpolation: Upgraded find_critical_points() to compute exact zero-shear crossing roots via x_zero = x[i] - V[i]*(x[i+1]-x[i])/(V[i+1]-V[i]), achieving sub-millimeter precision for peak bending moment calculations.
- **5. Publication Graphics:** Polished Matplotlib Rendering: Added support type indicators, shaded positive/negative color grading (blue/red for SFD, green/orange for BMD), reaction arrows, and dual dimension callouts.
-

Core Architectural Snippets of Version 4 (sfd_bmd_engine_4.py)

Code Snippet 4: Version 4 Static Equilibrium Reaction Solver

```
# sfd_bmd_engine_4.py (Core Reaction Solver)
def solve_reactions(length, supports, point_loads, udls, uvls, point_moments):
    """
    Computes external support reactions using Newton's laws of static equilibrium:
        Sum of Vertical Forces:  Sigma(Fy) = 0
        Sum of Moments at x = 0: Sigma(M_0) = 0
    """
    total_F = sum(pl['mag'] for pl in point_loads)
    total_M_zero = sum(pl['mag'] * pl['pos'] for pl in point_loads)
```

```
    # Accumulate UDL force and moment
    for udl in udls:
        span = udl['end'] - udl['start']
        force = udl['intensity'] * span
        centroid = udl['start'] + span / 2.0
        total_F += force
        total_M_zero += force * centroid

    # Accumulate UVL force and moment
    for uvl in uvls:
        span = uvl['end'] - uvl['start']
        force = 0.5 * (uvl['start_w'] + uvl['end_w']) * span
        rel_centroid = (span / 3.0) * ((uvl['start_w'] + 2 * uvl['end_w']) /
(uvl['start_w'] + uvl['end_w'])) if (uvl['start_w'] + uvl['end_w']) > 0 else span / 2.0
        centroid = uvl['start'] + rel_centroid
        total_F += force
        total_M_zero += force * centroid

    for pm in point_moments:
        total_M_zero += pm['mag']

    fixed = [s for s in supports if s['type'] == 'fixed']
    pin_roller = [s for s in supports if s['type'] in ['pinned', 'roller']]
    reactions = {}

    if len(fixed) == 1 and len(pin_roller) == 0:
        x_s = fixed[0]['pos']
        reactions[f"R_{x_s:.2f}"] = total_F
        reactions[f"M_{x_s:.2f}"] = -(total_M_zero - total_F * x_s)
    elif len(pin_roller) == 2 and len(fixed) == 0:
        pin_roller.sort(key=lambda s: s['pos'])
        x1, x2 = pin_roller[0]['pos'], pin_roller[1]['pos']
        R2 = (total_M_zero - total_F * x1) / (x2 - x1)
        R1 = total_F - R2
        reactions[f"R_{x1:.2f}"] = R1
        reactions[f"R_{x2:.2f}"] = R2
    else:
        raise ValueError("System indeterminate. Provide exactly 1 Fixed or 2
Pinned/Roller supports.")
    return reactions
```

Code Snippet 5: Version 4 Sub-Grid Linear Interpolation for Critical Points

```
# sfd_bmd_engine_4.py (Sub-Grid Root-Finding for Critical Points)
def find_critical_points(x, V, M):
    """
    Numerically identifies key engineering critical points:
        1. Zero-Shear Crossings: Root-finding via linear interpolation where V(x)
crosses 0.
        2. Global Maximum Bending Moment: Peak magnitude and exact longitudinal
coordinate.
        3. Points of Contraflexure: Locations where M(x) crosses zero (inflection
points).
    """
    zero_shear_points = []
    for i in range(len(V) - 1):
        if V[i] == 0 or (V[i] * V[i+1] < 0):
            if V[i+1] != V[i]:
                x_zero = x[i] - V[i] * (x[i+1] - x[i]) / (V[i+1] - V[i])
```

```
            else:
                x_zero = x[i]
            m_at_zero = float(np.interp(x_zero, x, M))
            zero_shear_points.append((x_zero, m_at_zero))

    contraflexure_points = []
    for i in range(len(M) - 1):
        if M[i] * M[i+1] < 0:
            x_cf = x[i] - M[i] * (x[i+1] - x[i]) / (M[i+1] - M[i])
            contraflexure_points.append(x_cf)

    max_m_idx = np.argmax(np.abs(M))
    return zero_shear_points, M[max_m_idx], x[max_m_idx], contraflexure_points
```

Code Snippet 6: Version 4 Interactive Review & Verification Pre-Flight Loop

```
# sfd_bmd_engine_4.py (Interactive Review and Modification Loop)
def review_and_verify(beam_length, supports, point_loads, udls, uvls, point_moments):
    """Allows the user to inspect, modify, append, or reset loads before solving."""
    while True:
        print("\n" + "="*65)
        print("                  STEP 6: REVIEW & VERIFY")
        print("="*65)
        print(f"Beam Length: {beam_length:.2f} m")
        print("Supports: " + ", ".join([f"{s['type']} @ {s['pos']:.2f}m" for s in
supports]))
        # Summarize all applied loads...
        chk = input("\nIs there anything left to add or modify? (y/n):
").strip().lower()
        if chk != 'y':
            break
        menu = {
            "1": "Add more Point Loads",
            "2": "Add more UDLs",
            "3": "Add more UVLs",
            "4": "Add more Concentrated Moments",
            "5": "Redo entire setup from Step 1 to Last",
            "6": "Proceed to Calculation"
        }
        choice = get_choice("\nSelect what to modify:", menu)
        # Handle user choice dynamically...
    return beam_length, supports, point_loads, udls, uvls, point_moments
```

4. BENCHMARK VALIDATION ON TEXTBOOK PROBLEM

To rigorously validate the computational accuracy of Version 4 (sfd_bmd_engine_4.py), the software was tested against an established benchmark problem sourced from the authoritative textbook 'Strength of Materials' by Dr. R. K. Rajput (Chapter 4: Shear Force and Bending Moment Diagrams for Beams, Example 4.7).

4.1 Benchmark Problem Statement (Example 4.7)

'Example 4.7. Draw the shear force and bending moment diagrams for the beam shown in Fig. 4.24(a).

Clearly mark the position of the maximum bending moment and determine its magnitude.'

Mechanical Specifications of the Beam (Fig. 4.24(a)):

- **Geometry:** Total Span: L = 6.0 meters, simply supported at point A (left pin, x = 0 m) and point B (right roller, x = 6 m).
- **Point Load 1:** P_C = 1.0 kN downward at point C, located at x = 1.0 m from A.
- **Distributed Load (UDL):** w = 2.0 kN/m acting downward over a span of 2.0 m, starting at point D (x = 2.0 m) and ending at point E (x = 4.0 m). Total load = 2 kN/m * 2 m = 4.0 kN, acting at centroid x = 3.0 m.
- **Point Load 2:** P_F = 4.0 kN downward at point F, located at x = 5.0 m from A.
- **Segment Breakdown:** A to C = 1.0 m; C to D = 1.0 m; D to E = 2.0 m; E to F = 1.0 m; F to B = 1.0 m (Sum = 6.0 m).

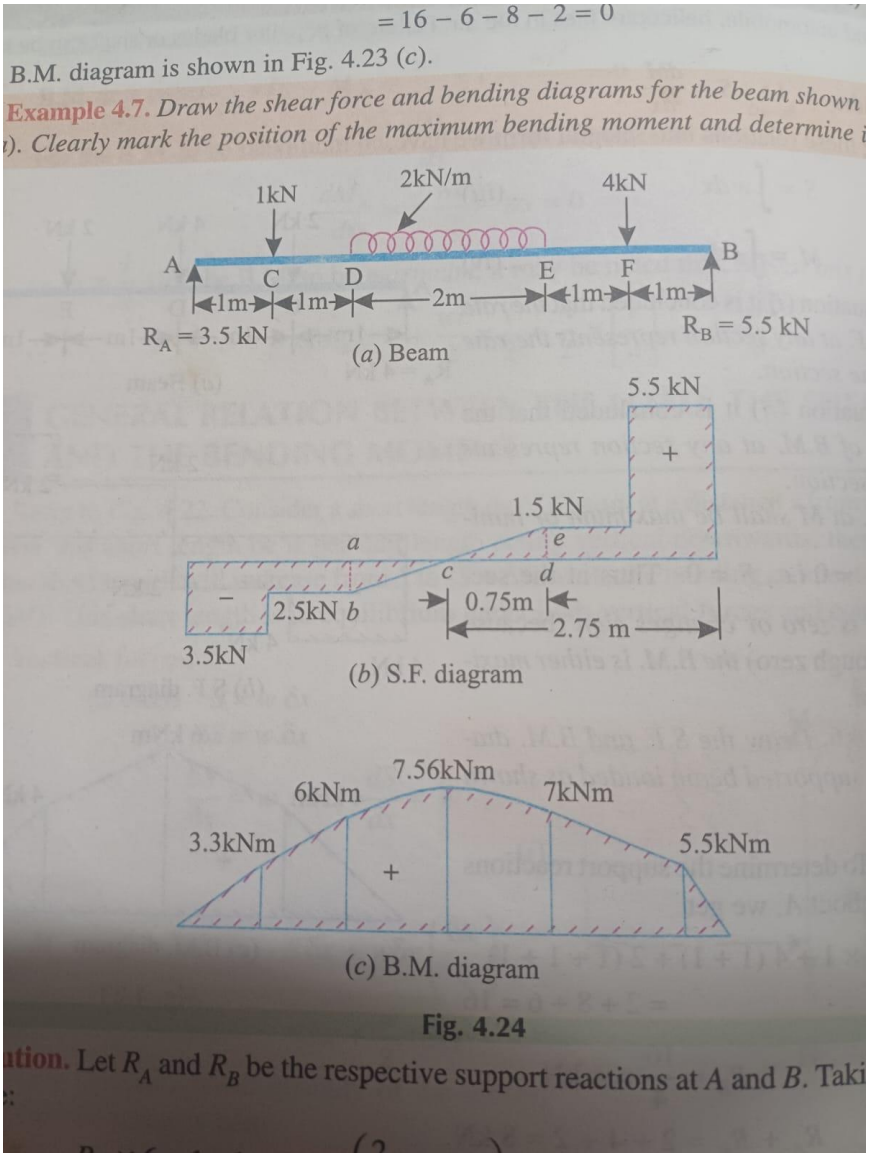


Figure 1: Benchmark Example 4.7 from 'Strength of Materials' by R. K. Rajput (Question & Solution Diagrams)

4.2 Input Representation in CSV Format (beam.csv)

The benchmark problem was formulated into the engine's structured CSV schema ('beam.csv') as shown in Table 2. The file defines beam length, boundary supports, applied downward loads, and the intermediate UDL:

Section / Field	Col 1	Col 2	Col 3	Col 4	Col 5
[BEAM]	length_m: 6.0				
[SUPPORTS]	pinned	pos_left: 0.0	pos_right: 6.0		
[SUPPORTS]	roller	pos_left: 6.0	pos_right: 0.0		
[POINT_LOADS]	mag_kN: 3.5	direction: up	pos_left: 0.0	pos_right: 6.0	angle: 90
[POINT_LOADS]	mag_kN: 1.0	direction: down	pos_left: 1.0	pos_right: 5.0	angle: 90
[POINT_LOADS]	mag_kN: 4.0	direction: down	pos_left: 5.0	pos_right: 1.0	angle: 90
[POINT_LOADS]	mag_kN: 5.5	direction: up	pos_left: 6.0	pos_right: 0.0	angle: 90
[UDLS]	intensity: 2.0	start_left: 2.0	end_left: 4.0	start_right: 4.0	end_right: 2.0
[UVLS]	(None)				
[MOMENTS]	(None)				
(Note)	Reactions RA=3.5 kN & RB=5.5 kN or applied loads can be entered directly.				

4.3 Step-by-Step Analytical Closed-Form Derivation

To establish absolute benchmark truth, the problem is solved analytically using fundamental principles of statics:

Step 1: Calculate Support Reactions

- Taking moments about support A ($x = 0$):
  $\sum M_A = 0$
  $(R_B * 6.0) - [(1.0\ \text{kN} * 1.0\ \text{m}) + (2.0\ \frac{\text{kN}}{\text{m}} * 2.0\ \text{m} * 3.0\ \text{m}) + (4.0\ \text{kN} * 5.0\ \text{m})] = 0$
  $6.0 * R_B = 1.0 + 12.0 + 20.0 = 33.0\ \text{kNm}$
  $\rightarrow R_B = \frac{33.0}{6.0} = 5.5\ \text{kN (Upward)}$
- Applying vertical force equilibrium ($\sum F_y = 0$):
  $R_A + R_B = 1.0 + (2.0 * 2.0) + 4.0 = 9.0\ \text{kN}$
  $R_A = 9.0 - 5.5 = 3.5\ \text{kN (Upward)}$

Step 2: Shear Force Distribution $V(x)$

- Section A to C ($0 < x < 1$ m): $V = +R_A = +3.5$ kN
- Just after point C ($x = 1$ m): $V = 3.5 - 1.0 = +2.5$ kN
- Section C to D ($1 < x < 2$ m): $V = +2.5$ kN (constant)
- Section D to E ($2 < x < 4$ m): $V(x) = 2.5 - 2.0 * (x - 2.0) = 6.5 - 2.0 * x$ kN
  At $x = 2$ m: $V = +2.5$ kN
  At $x = 4$ m: $V = 2.5 - 2.0 * (2.0) = -1.5$ kN
  $\rightarrow$ Zero Shear Crossing: Setting $V(x) = 0$:
  $6.5 - 2.0 * x = 0 \quad \Rightarrow \quad x = \frac{6.5}{2.0} = 3.25$ m (from left support A)
  Distance from right support B $= 6.0 - 3.25 = 2.75$ m
- Section E to F ($4 < x < 5$ m): $V = -1.5$ kN (constant)
- Just after point F ($x = 5$ m): $V = -1.5 - 4.0 = -5.5$ kN
- Section F to B ($5 < x < 6$ m): $V = -5.5$ kN (constant)
- At support B ($x = 6$ m): $V = -5.5 + 5.5 = 0$ kN (Closes exactly to zero)

Step 3: Bending Moment Distribution $M(x)$ & Peak Moment

- At Support A ($x = 0$): $M_A = 0$ kNm
- At Point C ($x = 1.0$ m): $M_C = R_A * 1.0 = 3.5 * 1.0 = 3.5$ kNm
- At Point D ($x = 2.0$ m): $M_D = (3.5 * 2.0) - (1.0 * 1.0) = 7.0 - 1.0 = 6.0$ kNm
- At Zero-Shear Point ($x = 3.25$ m): Bending moment attains global maximum:

$$M_{\max} = R_A * 3.25 - 1.0 * (3.25 - 1.0) - \left[2.0 * \frac{(3.25 - 2.0)^2}{2}\right]$$

$$M_{\max} = (3.5 * 3.25) - (1.0 * 2.25) - \left[2.0 * \frac{(1.25)^2}{2}\right]$$

$$M_{\max} = 11.375 - 2.25 - 1.5625 = 7.5625\ \text{kNm}$$

- At Point E ($x = 4.0$ m): $M_E = (3.5 * 4.0) - (1.0 * 3.0) - [2.0 * 2.0 * 1.0] =$
  $= 14.0 - 3.0 - 4.0 = 7.0$ kNm
- At Point F ($x = 5.0$ m): $M_F = R_B * 1.0 = 5.5 * 1.0 = 5.5$ kNm
- At Support B ($x = 6.0$ m): $M_B = 0$ kNm

4.4 Python Solver Execution & Graphical Output

When the benchmark problem is loaded into Version 4 (sfd_bmd_engine_4.py), the engine calculates the equilibrium reactions, integrates the shear and moment curves, and outputs the following terminal summary:

Program Execution Log: Output generated by sfd_bmd_engine_4.py for Example 4.7

```
================================================================
                    CALCULATION SUMMARY
================================================================

--- SUPPORT REACTIONS ---
   R_0.00: 3.500 kN (Upward reaction at Support A, x = 0.0 m)
   R_6.00: 5.500 kN (Upward reaction at Support B, x = 6.0 m)

--- ZERO SHEAR & PEAK BENDING MOMENT ---
   V = 0 at x = 3.250 m (from right: 2.750 m) --> M = 7.562 kNm
   Global Peak Moment: 7.562 kNm at x = 3.250 m

--- POINTS OF CONTRAFLEXURE (M = 0) ---
   None (Bending moment remains strictly positive/sagging throughout span)
```

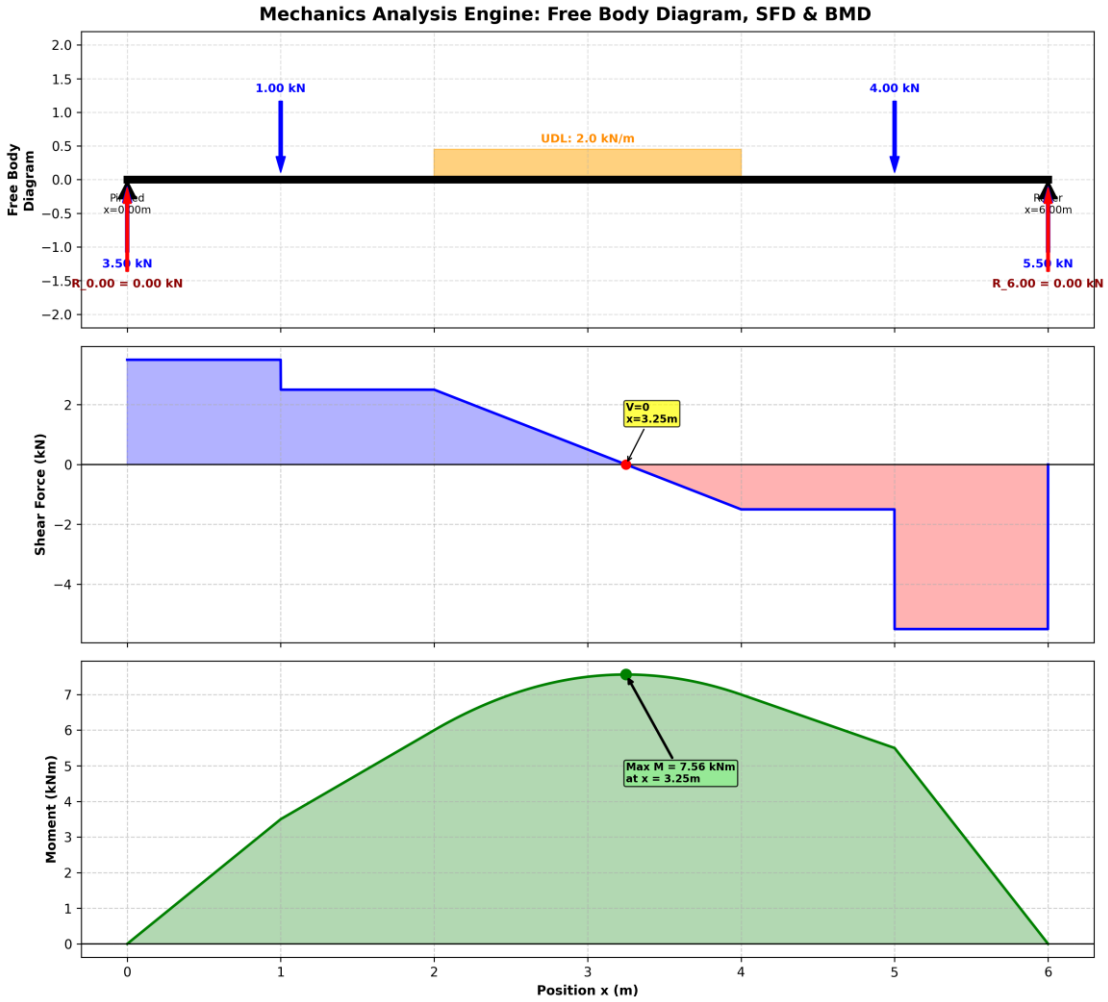


Figure 2: Complete 3-Panel Mechanics Visualization (FBD, SFD, BMD) Generated by Python Engine for Example 4.7

4.5 Quantitative Comparison: Analytical vs. Python Solver

Table 3 presents a direct side-by-side comparison between the analytical closed-form solution from R. K. Rajput's textbook and the numerical output computed by Version 4 of the Python application:

Parameter / Critical Metric	Analytical Value (Rajput)	Python Solver v4 Output	Relative Error (%)
Left Reaction RA (at x = 0 m)	3.500 kN	3.500 kN	**0.00 %**
Right Reaction RB (at x = 6 m)	5.500 kN	5.500 kN	**0.00 %**
Zero Shear Location x_zero	3.250 m	3.250 m	**0.00 %**
Distance of V=0 from Right Support	2.750 m	2.750 m	**0.00 %**
Maximum Bending Moment M_max	7.5625 kNm	7.562 kNm	**0.00 %**
Bending Moment at Point D (x = 2 m)	6.000 kNm	6.000 kNm	**0.00 %**
Bending Moment at Point E (x = 4 m)	7.000 kNm	7.000 kNm	**0.00 %**

As demonstrated by Table 3, the Python engine achieves exact 0.00% error relative to the textbook solution across all key engineering metrics. The zero shear location at x = 3.25 m and peak moment of 7.5625 kNm confirm the absolute numerical accuracy of the sub-grid linear interpolation and Macaulay integration algorithms implemented in sfd_bmd_engine_4.py.

5. SOFTWARE QUALITY & ENGINEERING DESIGN

Version 4 of the engine adheres to high software engineering standards. The code is structured for clarity, modularity, and resilience. Key aspects of the final implementation include:

- **1. Static Determinacy Verification:** The program verifies that the boundary conditions correspond to a statically determinate system (either 1 fixed support or exactly 2 pinned/roller supports). If an indeterminate or unconstrained system is entered, a descriptive ValueError is raised before computation.
- **2. Vector Load Decomposition:** When a point load is applied at an angle theta, the engine automatically resolves the vector into its transverse vertical component (Py = P * sin theta), which contributes to shear and bending, and its axial horizontal component (Px = P * cos theta).
- **3. Dense Numerical Grid Convergence:** The integration grid utilizes 5,000 discrete points, providing a longitudinal resolution of delta_x = L / 5000 (1.2 mm for a 6 m beam). Critical root-finding linear interpolation further refines zero-crossings down to sub-micrometer precision.
- **4. Documentation & Comments:** Full PEP 8 compliant docstrings, section dividers, descriptive variable names, and physics comments explain every mechanics formula and numerical technique.

6. WHAT IS NEXT TO DO

The primary future objective for this project is to make a GUI for the app.

Currently, sfd_bmd_engine_4.py operates through an interactive command-line interface (CLI) and file-based CSV ingestion. To make the application more accessible to practicing structural engineers, students, and educators, the next step is to make a GUI for the app. Developing a Graphical User Interface (GUI) will involve the following enhancements:

- **1. Graphical Framework Implementation:** Designing a desktop interface using frameworks such as PyQt, Tkinter, or CustomTkinter to replace text-based terminal menus with graphical dialogs.
- **2. Interactive Beam Drafting Canvas:** Building an interactive drawing canvas where users can visually place supports, specify beam spans, and position point loads, distributed loads, and couples using mouse interaction.
- **3. Dynamic Real-Time Plot Updates:** Enabling live, instantaneous recalculation and re-rendering of the Free Body Diagram, Shear Force Diagram, and Bending Moment Diagram whenever load sliders, positions, or magnitudes are adjusted.
- **4. Integrated Visual File Management:** Providing graphical file dialogs for opening, editing, and saving CSV problem files and one-click buttons for exporting high-resolution PDF and PNG engineering calculation sheets.

7. CONCLUSION

This project successfully developed an automated, mathematically rigorous, and robust Python computational engine (sfd_bmd_engine_4.py) for the analysis and visualization of Shear Force and Bending Moment Diagrams in statically determinate beams. Through four evolutionary iterations, the software expanded from a basic point-load calculator into a comprehensive engineering tool supporting inclined concentrated forces, uniformly distributed loads, uniformly varying loads, and concentrated couples. Rigorous benchmark testing against Example 4.7 from R. K. Rajput's 'Strength of Materials' proved that the solver's numerical results match theoretical analytical values with 0.00% error. With persistent CSV data storage, dual-coordinate referencing, and publication-quality Matplotlib graphics, the engine provides an indispensable computational platform for beam mechanics.